Noah Sun

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Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) September 2025 – April 2030

Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic GPA: 3.92 / 4.00

Experience

Full Stack Software Engineer – Thames Valley Financial September 2026 – Current

- Replaced a legacy Excel model with **3 actuarial calculators** on a Python engine that runs **360+ scenarios/second**
- Designed a daily fund-trading pipeline for a retirement plan by building a Python model, validated by **87 test cases**
- Built an interactive compliance engine demo of **12 scenarios** across **8 flags**, used to show its versatility and uses

Humanoid Perception Software Developer – WATonomous September 2026 – Current

- Built a ROS2 pipeline that localizes a shuttlecock from RGB-D data using YOLO detection and depth back-projection

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers to build a 32 MB C++ flash storage layer with integrity checks, verified by **19 GoogleTest** unit tests
- Cut flash I/O CPU time **15%** by building a non-blocking, RTOS-integrated state machine with async queues

Forward Deployed Engineer – Ascendance Foundry January 2026 – April 2026

- Cut process errors by **80% (50 to 9 per month)** by building **5 AI workflow automations** with input validation
- Saved **6 hours per week** by automating lead discovery and outreach drafts with Python Playwright and Claude API
- Exposed spending inefficiencies across **50+ projects** by building **2 Azure ETL pipelines** feeding Power BI dashboards

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub June 2024 – August 2025

- Reached **3500+ downloads** and a **5.0-star rating** by leading 4 developers to ship a Flutter app for VEX tournaments
- Reduced scouting times by **30%** by syncing team data with Firebase and Cloud Functions and predictive match times

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub June 2019 – August 2025

- Won Create Award at VEX Worlds 2024 among **82 teams** by leading 6 members in robot design and library architecture
- Qualified to VEX Worlds **5 straight years** by building ArkLib, a modular C++ motion library **reused across 16 robots**

Projects

Signapse | Live Demo | GitHub

- Co-built an ASL translator for clinical settings integrating computer vision, Gemini language layer, and TimescaleDB
- Implemented live recognition of **9 signs** from **92 examples** by applying DTW + k-NN to MediaPipe hand landmarks

Redact | Live Demo | GitHub

- Built a full-stack face-blurring app (FastAPI, React, Docker) combining RetinaFace detection with OpenCV tracking
- Sped up video processing **5x to 4 FPS** with **97% face recall** by pairing RetinaFace every 5th frame with IoU tracking
- Reduced blur-box jitter by **30%** across shot changes by designing dual-signal scene cut detection and EMA smoothing

Signal - LLM Agent Simulation Platform | GitHub

- Built **3-layer** simulation engine, Claude tool-use harness, and FastAPI/SSE server with live results on a dashboard
- Cut LLM cost per call **55%** ($0.0236 to $0.0106) with no drop in test scores by implementing prompt caching

CityWatch | GitHub

- Led a team of 4 to build a reporting platform using React Native to analyze **50k+ reports** to identify high-risk zones
- Classified road defects at **0.95 macro F1** on **1603 images** by training MobileNetV2 with OpenCV preprocessing
- Forecasted 30-365 day risk heatmaps from spatial incident patterns and features stored in a PostGIS database

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart
Frontend & Mobile: React, React Native, Flutter, HTML, CSS, Expo
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, SSE, Firebase Cloud Functions, Playwright
AI & Computer Vision: OpenCV, TensorFlow, MediaPipe, RetinaFace, MobileNetV2, DTW, k-NN, Claude, Gemini
Data & Infrastructure: PostgreSQL, MySQL, PostGIS, TimescaleDB, Firebase, Azure, Power BI, Pandas, Docker, Git, Linux